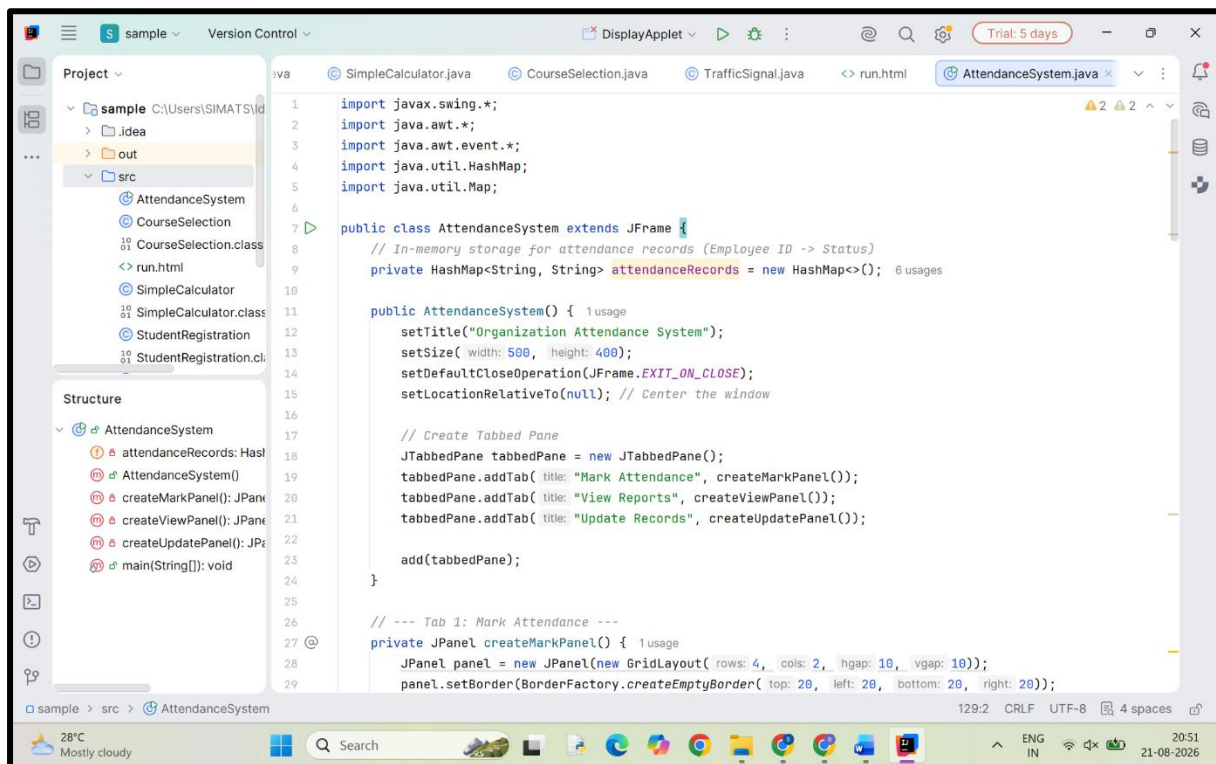


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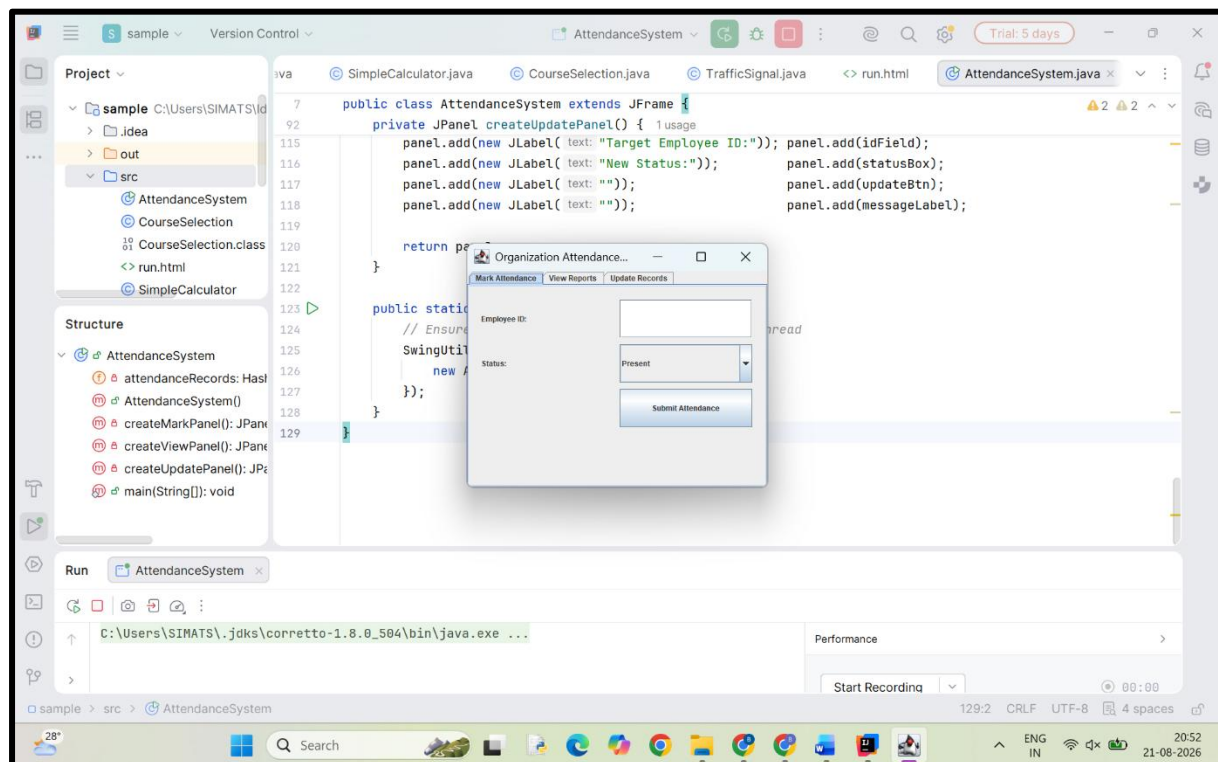
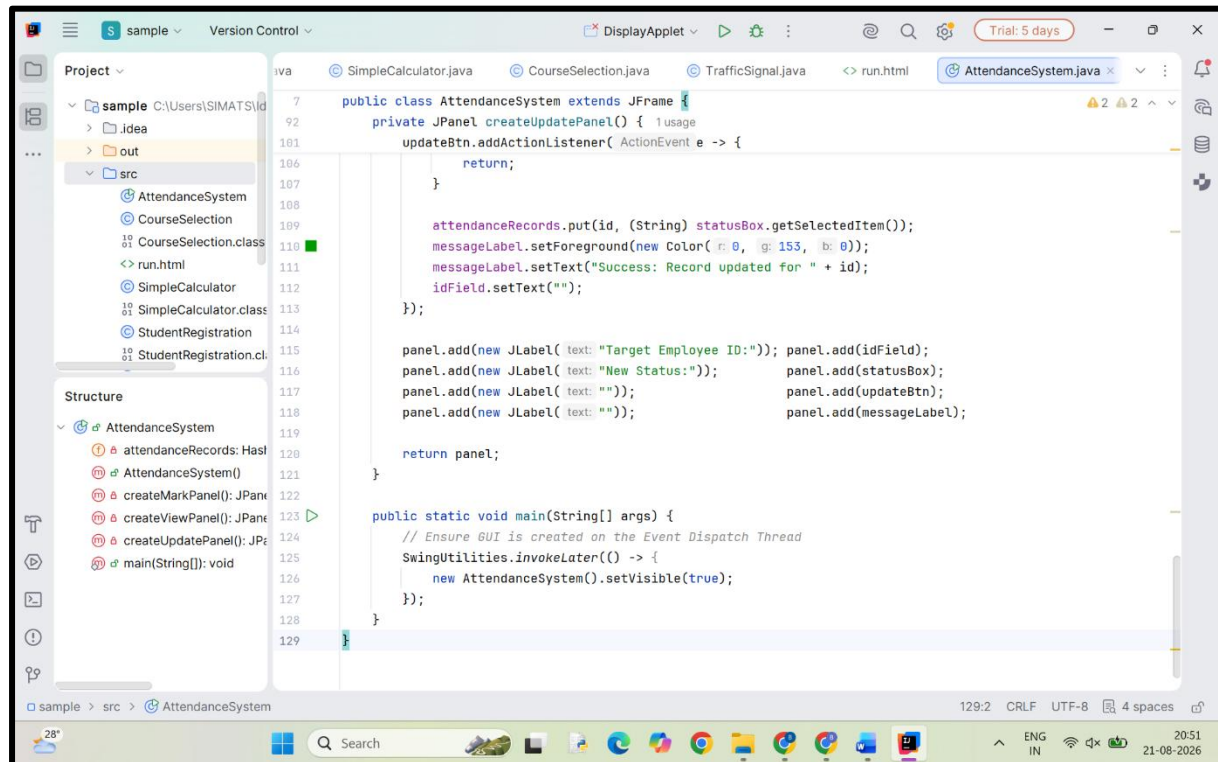
CO4 AT1 INDUSTRY PROBLEM SOLVING

Develop a GUI-based attendance system for an organization using Swing components. The system should allow marking attendance, viewing reports, and updating records. Use event handling to process user inputs and actions. Implement validation to prevent duplicate entries and incorrect data. Apply suitable layout managers for structured display. Analyze system responsiveness during bulk data entry and suggest enhancements for better performance and usability.

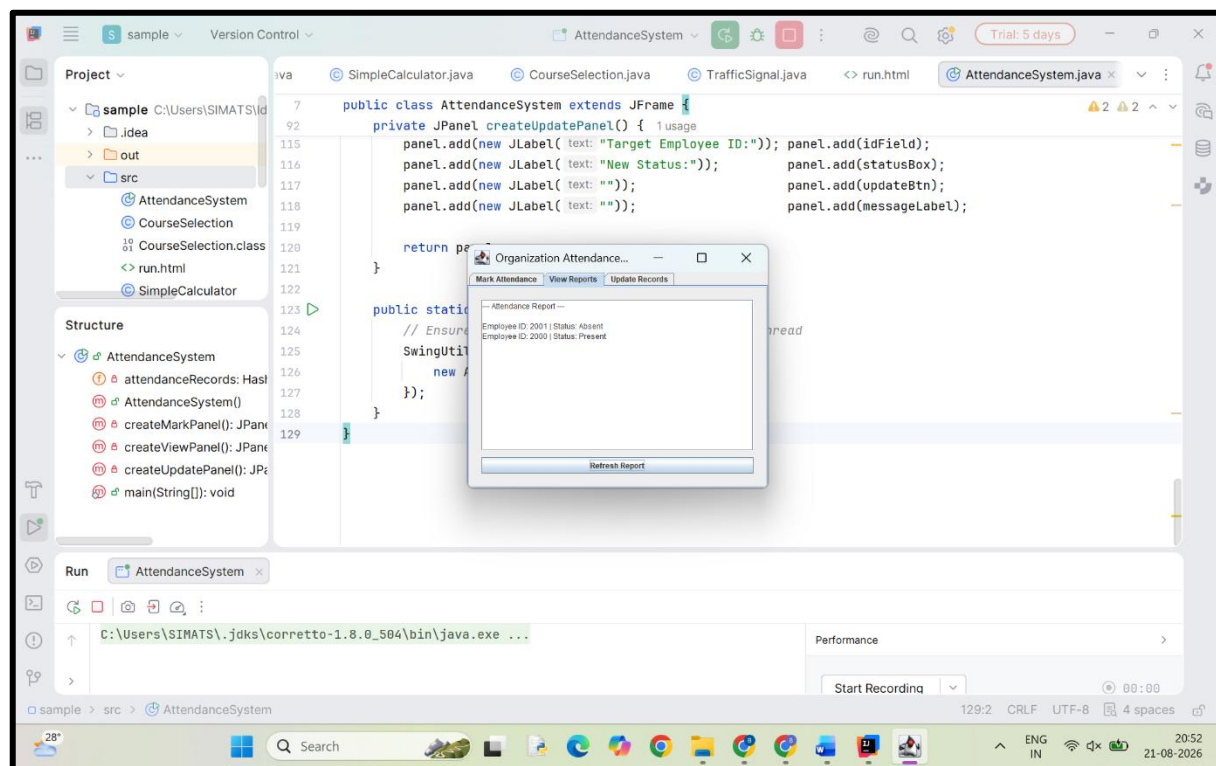
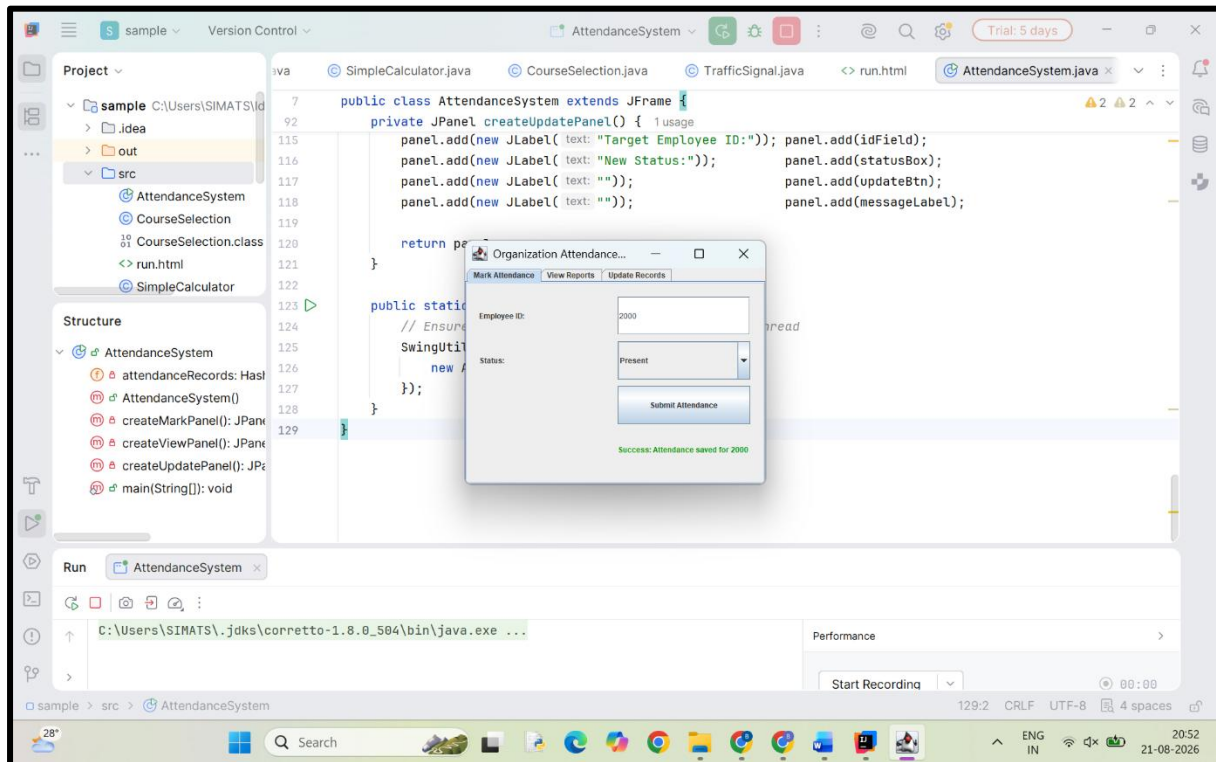


```
1 import javax.swing.*;
2 import java.awt.*;
3 import java.awt.event.*;
4 import java.util.HashMap;
5 import java.util.Map;
6
7 public class AttendanceSystem extends JFrame {
8     // In-memory storage for attendance records (Employee ID -> Status)
9     private HashMap<String, String> attendanceRecords = new HashMap<>();
10
11     public AttendanceSystem() {
12         setTitle("Organization Attendance System");
13         setSize( width: 500, height: 400);
14         setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
15         setLocationRelativeTo(null); // Center the window
16
17         // Create Tabbed Pane
18         JTabbedPane tabbedPane = new JTabbedPane();
19         tabbedPane.addTab( title: "Mark Attendance", createMarkPanel());
20         tabbedPane.addTab( title: "View Reports", createViewPanel());
21         tabbedPane.addTab( title: "Update Records", createUpdatePanel());
22
23         add(tabbedPane);
24     }
25
26     // --- Tab 1: Mark Attendance ---
27     private JPanel createMarkPanel() {
28         JPanel panel = new JPanel(new GridLayout( rows: 4, cols: 2, hgap: 10, vgap: 10));
29         panel.setBorder(BorderFactory.createEmptyBorder( top: 20, left: 20, bottom: 20, right: 20));
```

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